Experiment 5(a): Masking and Setting of Lower Nibbles Using 8085

AIM:

To write and execute an assembly language program for performing masking and setting of lower nibbles on given 8-bit data using the 8085 microprocessor.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

(i) Masking of Lower Nibble

ALGORITHM:

1. Load the data from memory location 4200H into the accumulator.
2. Perform logical AND with 0FH.
3. Store the result in memory location 4201H.
4. Stop the program.

PROGRAM:

LDA 4200H

ANI 0FH

STA 4201H

HLT

(ii) Setting of Lower Nibble

ALGORITHM:

1. Load the data from memory location 4200H into the accumulator.
2. Perform logical OR with 0FH.
3. Store the result in memory location 4201H.
4. Stop the program.

PROGRAM:

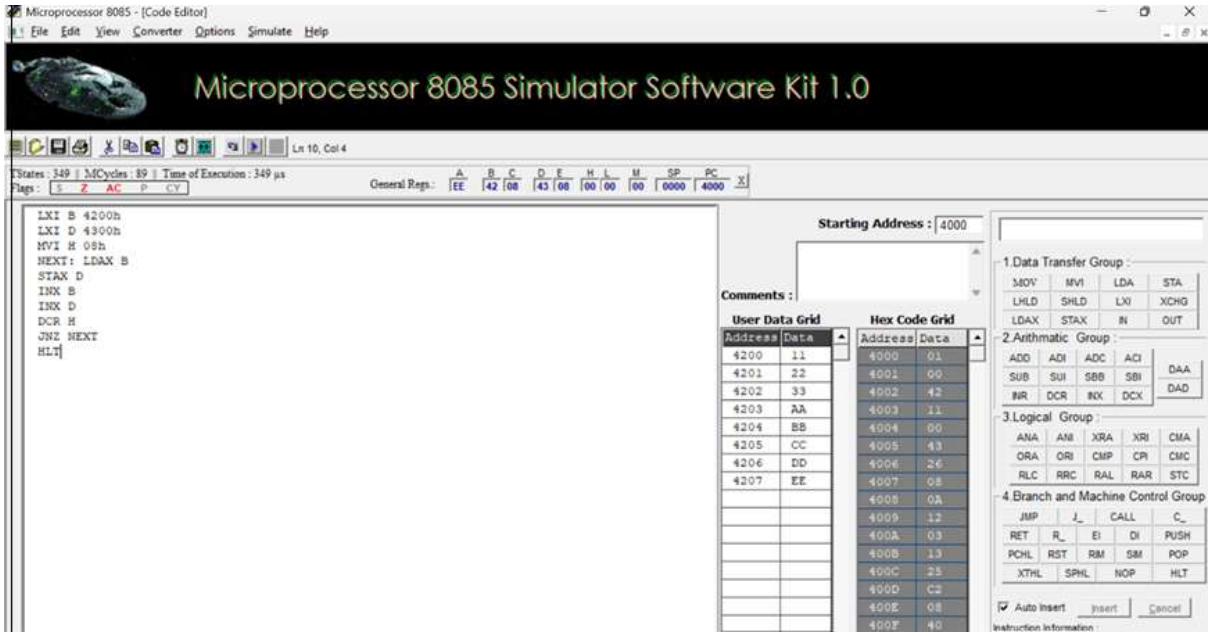
LDA 4200H

ORI 0FH

STA 4201H

HLT

OUTPUT:



RESULT:

Thus, the assembly language programs for masking and setting the lower nibble of 8-bit data were written and executed successfully using the 8085 microprocessor.